APPENDIX II
PROOFS FROM SECTION V

A. *Proof for Lemma 1*

Proof. Let $\mathbf{x}^* \in \mathcal{X}^*$. Then, for all $t \geq 0$,

$$\mathbf{x}_{t+1} - \mathbf{x}^* = \mathbf{x}_t - \mathbf{x}^* + \beta_t(\mathbf{v}(\mathbf{x}_t) - \mathbf{v}(\mathbf{x}^*) + \mathbf{M}_{t+1}).$$

Now, using the standard expansion, we get

$$\begin{aligned}\|\mathbf{x}_{t+1} - \mathbf{x}^*\|^2 &= \|\mathbf{x}_t - \mathbf{x}^* + \beta_t(\mathbf{v}(\mathbf{x}_t) - \mathbf{v}(\mathbf{x}^*) + \mathbf{M}_{t+1})\|^2 \\ &= \|\mathbf{x}_t - \mathbf{x}^*\|^2 + \beta_t^2\|\mathbf{v}(\mathbf{x}_t) - \mathbf{v}(\mathbf{x}^*) + \mathbf{M}_{t+1}\|^2 \qquad \text{(6a)} \\ &\quad + 2\beta_t\langle \mathbf{x}_t - \mathbf{x}^*, \mathbf{v}(\mathbf{x}_t) - \mathbf{v}(\mathbf{x}^*) + \mathbf{M}_{t+1}\rangle. \qquad \text{(6b)}\end{aligned}$$

For the term (6a), we note that

$$\|\mathbf{v}(\mathbf{x}_t) - \mathbf{v}(\mathbf{x}^*) + \mathbf{M}_{t+1}\|^2 = \|\mathbf{v}(\mathbf{x}_t) - \mathbf{v}(\mathbf{x}^*)\|^2 + \|\mathbf{M}_{t+1}\|^2 + 2\langle \mathbf{v}(\mathbf{x}_t) - \mathbf{v}(\mathbf{x}^*), \mathbf{M}_{t+1}\rangle.$$

For the term (6b), since $-\mathbf{v}(\cdot)$ is λ-co-coercive, we have

$$\langle \mathbf{x}_t - \mathbf{x}^*, \mathbf{v}(\mathbf{x}_t) - \mathbf{v}(\mathbf{x}^*)\rangle \leq -\lambda\|\mathbf{v}(\mathbf{x}_t) - \mathbf{v}(\mathbf{x}^*)\|^2.$$

Hence,

$$2\beta_t\langle \mathbf{x}_t - \mathbf{x}^*, \mathbf{v}(\mathbf{x}_t) - \mathbf{v}(\mathbf{x}^*) + \mathbf{M}_{t+1}\rangle \leq -2\lambda\beta_t\|\mathbf{v}(\mathbf{x}_t) - \mathbf{v}(\mathbf{x}^*)\|^2 + 2\beta_t\langle \mathbf{x}_t - \mathbf{x}^*, \mathbf{M}_{t+1}\rangle.$$

Returning to (6), we get

$$\begin{aligned}\|\mathbf{x}_{t+1} - \mathbf{x}^*\|^2 &\leq \|\mathbf{x}_t - \mathbf{x}^*\|^2 + \beta_t^2\|\mathbf{v}(\mathbf{x}_t) - \mathbf{v}(\mathbf{x}^*)\|^2 + \beta_t^2\|\mathbf{M}_{t+1}\|^2 \\ &\quad + 2\beta_t^2\langle \mathbf{v}(\mathbf{x}_t) - \mathbf{v}(\mathbf{x}^*), \mathbf{M}_{t+1}\rangle - 2\lambda\beta_t\|\mathbf{v}(\mathbf{x}_t) - \mathbf{v}(\mathbf{x}^*)\|^2 + 2\beta_t\langle \mathbf{x}_t - \mathbf{x}^*, \mathbf{M}_{t+1}\rangle \\ &= \|\mathbf{x}_t - \mathbf{x}^*\|^2 - \beta_t(2\lambda - \beta_t)\|\mathbf{v}(\mathbf{x}_t) - \mathbf{v}(\mathbf{x}^*)\|^2 + \beta_t^2\|\mathbf{M}_{t+1}\|^2 \\ &\quad + 2\beta_t\langle \mathbf{x}_t - \mathbf{x}^*, \mathbf{M}_{t+1}\rangle + 2\beta_t^2\langle \mathbf{v}(\mathbf{x}_t) - \mathbf{v}(\mathbf{x}^*), \mathbf{M}_{t+1}\rangle \\ &\leq \|\mathbf{x}_t - \mathbf{x}^*\|^2 - \lambda\beta_t\|\mathbf{v}(\mathbf{x}_t) - \mathbf{v}(\mathbf{x}^*)\|^2 + \beta_t^2\|\mathbf{M}_{t+1}\|^2 \\ &\quad + 2\beta_t\langle \mathbf{x}_t - \mathbf{x}^*, \mathbf{M}_{t+1}\rangle + 2\beta_t^2\langle \mathbf{v}(\mathbf{x}_t) - \mathbf{v}(\mathbf{x}^*), \mathbf{M}_{t+1}\rangle.\end{aligned}$$

Here the final inequality follows from the fact that $\beta_t \leq \lambda$ which implies that $2\lambda - \beta_t \geq \lambda$ for all t. On taking expectation conditioned on $\mathcal{F}_t$, we have

$$\mathbb{E}[\langle \mathbf{x}_t - \mathbf{x}^*, \mathbf{M}_{t+1}\rangle \mid \mathcal{F}_t] = 0, \qquad \mathbb{E}[\langle \mathbf{v}(\mathbf{x}_t) - \mathbf{v}(\mathbf{x}^*), \mathbf{M}_{t+1}\rangle \mid \mathcal{F}_t] = 0,$$

since $\{\mathbf{M}_{t+1}\}$ is a martingale difference sequence with respect to $\{\mathcal{F}_t\}$, and both $\mathbf{x}_t$ and $\mathbf{v}(\mathbf{x}_t)$ are $\mathcal{F}_t$-measurable. Therefore,

$$\begin{aligned}\mathbb{E}\left[\|\mathbf{x}_{t+1} - \mathbf{x}^*\|^2 \mid \mathcal{F}_t\right] &\leq \|\mathbf{x}_t - \mathbf{x}^*\|^2 + \beta_t^2\mathbb{E}\left[\|\mathbf{M}_{t+1}\|^2 \mid \mathcal{F}_t\right] - \lambda\beta_t\|\mathbf{v}(\mathbf{x}_t) - \mathbf{v}(\mathbf{x}^*)\|^2 \\ &\leq \|\mathbf{x}_t - \mathbf{x}^*\|^2 + \beta_t^2\sigma^2(1 + \|\mathbf{x}_t\|^2) - \lambda\beta_t\|\mathbf{v}(\mathbf{x}_t)\|^2 \\ &\leq \|\mathbf{x}_t - \mathbf{x}^*\|^2 + 2\beta_t^2\sigma^2\|\mathbf{x}_t - \mathbf{x}^*\|^2 + \beta_t^2\sigma^2(1 + 2\|\mathbf{x}^*\|^2) - \lambda\beta_t\|\mathbf{v}(\mathbf{x}_t)\|^2,\end{aligned}$$

where we used $\mathbf{v}(\mathbf{x}^*) = 0$ for the second inequality. This completes the proof for Lemma 1.

□

B. *Proof for Lemma 2*

Proof. Taking expectation and dropping the negative residual term in Lemma 1 gives, for all $t \geq 0$,

$$\mathbb{E}\left[\|\mathbf{x}_{t+1} - \mathbf{x}^*\|^2\right] \leq \left(1 + 2\beta_t^2\sigma^2\right)\mathbb{E}\left[\|\mathbf{x}_t - \mathbf{x}^*\|^2\right] + \beta_t^2\sigma^2(1 + 2\|\mathbf{x}^*\|^2).$$

Expanding the recursion, we get the following.

$$\begin{aligned}\mathbb{E}\left[\|\mathbf{x}_t - \mathbf{x}^*\|^2\right] &\leq \|\mathbf{x}_0 - \mathbf{x}^*\|^2\prod_{i=0}^{t-1}\left(1 + 2\beta_i^2\sigma^2\right) + \sum_{i=0}^{t-1}\sigma^2\beta_i^2(1 + 2\|\mathbf{x}^*\|^2)\prod_{j=i+1}^{t-1}\left(1 + 2\sigma^2\beta_j^2\right) \\ &\leq e^{2\sigma^2\sum_{i=0}^{\infty}\beta_i^2}\left(\|\mathbf{x}_0 - \mathbf{x}^*\|^2 + \sigma^2\left(1 + 2\|\mathbf{x}^*\|^2\right)\sum_{i=0}^{t-1}\beta_i^2\right)\end{aligned}$$

Let $D_1 \coloneqq \sum_{i=0}^{\infty}\beta_i^2 < \infty$. Then,

$$\mathbb{E}\left[\|\mathbf{x}_t - \mathbf{x}^*\|^2\right] \leq e^{2\sigma^2 D_1}\left(\|\mathbf{x}_0 - \mathbf{x}^*\|^2 + \sigma^2\left(1 + 2\|\mathbf{x}^*\|^2\right)D_1\right) =: \Gamma_1.$$

Hence $\mathbb{E}\left[\|\mathbf{x}_t - \mathbf{x}^*\|^2\right] \leq \Gamma_1$ for all $t \geq 0$. □

C. *Proof for Lemma 3*

Proof. Taking expectation in Lemma 1, and rearranging terms, we get

$$\lambda\beta_i\mathbb{E}\left[\|\mathbf{v}(\mathbf{x}_i)\|^2\right] \le \mathbb{E}\left[\|\mathbf{x}_i-\mathbf{x}^*\|^2\right] - \mathbb{E}\left[\|\mathbf{x}_{i+1}-\mathbf{x}^*\|^2\right] + \beta_i^2\sigma^2\big(1+2\|\mathbf{x}^*\|^2\big) + 2\beta_i^2\sigma^2\mathbb{E}\left[\|\mathbf{x}_i-\mathbf{x}^*\|^2\right].$$

Summing from $i=0$ to t gives

$$\lambda\sum_{i=0}^{t}\beta_i\mathbb{E}\left[\|\mathbf{v}(\mathbf{x}_i)\|^2\right] \le \mathbb{E}\left[\|\mathbf{x}_0-\mathbf{x}^*\|^2\right] - \mathbb{E}\left[\|\mathbf{x}_{t+1}-\mathbf{x}^*\|^2\right] + \sigma^2\big(1+2\|\mathbf{x}^*\|^2\big)\sum_{i=0}^{t}\beta_i^2 + 2\sigma^2\sum_{i=0}^{t}\beta_i^2\mathbb{E}\left[\|\mathbf{x}_i-\mathbf{x}^*\|^2\right].$$

Dropping the nonnegative term $\mathbb{E}[\|\mathbf{x}_{t+1}-\mathbf{x}^*\|^2]$, and using the fact that $\sum_t\beta_t^2 = D_1$ along with Lemma 2, we obtain

$$\lambda\sum_{i=0}^{t}\beta_i\mathbb{E}\left[\|\mathbf{v}(\mathbf{x}_i)\|^2\right] \le \mathbb{E}\left[\|\mathbf{x}_0-\mathbf{x}^*\|^2\right] + \sigma^2\big(1+2\|\mathbf{x}^*\|^2\big)D_1 + 2\sigma^2 D_1\Gamma_1.$$

This completes the proof for Lemma 3 with $\Gamma_2 := \frac{1}{\lambda}\left(\|\mathbf{x}_0-\mathbf{x}^*\|^2 + \sigma^2 D_1\left(1+2\|\mathbf{x}^*\|^2+2\Gamma_1\right)\right)$ □

D. *Proof for Lemma 4*

Proof. We first show that $\mathbf{v}$ is Lipschitz. For all $\mathbf{x},\mathbf{x}'\in\mathbb{R}^{Nd}$,

$$\begin{aligned}\lambda\|\mathbf{v}(\mathbf{x})-\mathbf{v}(\mathbf{x}')\|^2 &\le -\langle\mathbf{x}-\mathbf{x}',\mathbf{v}(\mathbf{x})-\mathbf{v}(\mathbf{x}')\rangle\\ &\le \|\mathbf{x}-\mathbf{x}'\|\|\mathbf{v}(\mathbf{x})-\mathbf{v}(\mathbf{x}')\|\\ \implies \|\mathbf{v}(\mathbf{x})-\mathbf{v}(\mathbf{x}')\| &\le \frac{1}{\lambda}\|\mathbf{x}-\mathbf{x}'\|. \end{aligned} \tag{7}$$

(a) Bound on $\mathbb{E}\left[\|\mathbf{U}_t\|^2\right]$. We first note that $\mathbb{E}[\|\mathbf{M}_{t+1}\|^2]$ is bounded by a constant for all $t\ge 0$.

$$\begin{aligned}\mathbb{E}\left[\|\mathbf{M}_{t+1}\|^2\right] &\le \sigma^2(1+\mathbb{E}\left[\|\mathbf{x}_t\|^2\right])\\ &\le \sigma^2(1+2\mathbb{E}\left[\|\mathbf{x}_t-\mathbf{x}^*\|^2\right]+2\|\mathbf{x}^*\|^2)\\ &\le \sigma^2(1+2\Gamma_1+2\|\mathbf{x}^*\|^2) =: D_2.\end{aligned}$$

Now, we note that

$$\mathbf{U}_t = \sum_{i=0}^{t-1}\beta_i\prod_{j=i+1}^{t-1}(1-\beta_j)\mathbf{M}_{i+1}.$$

Using the property that the terms of a martingale difference sequence are orthogonal,

$$\mathbb{E}\left[\|\mathbf{U}_t\|^2\right] \le \sum_{i=0}^{t-1}\left(\beta_i\prod_{j=i+1}^{t-1}(1-\beta_j)\right)^2\mathbb{E}\left[\|\mathbf{M}_{i+1}\|^2\right] \le D_2\sum_{i=0}^{t-1}\beta_i^2\prod_{j=i+1}^{t-1}(1-\beta_j)$$

Define $S_t = \sum_{i=0}^{t-1}\beta_i^2\prod_{j=i+1}^{t-1}(1-\beta_j)$. We next show that $S_t\le 2\beta_t$ for all $t\ge 0$. For this, we first note that,

$$\begin{aligned}\beta_t-\beta_{t+1} &= \frac{1}{(t+T_0)^{\mathfrak{b}}} - \frac{1}{(t+1+T_0)^{\mathfrak{b}}}\\ &= \frac{1}{(t+T_0)^{\mathfrak{b}}}\left(1-\left(1+\frac{1}{t+T_0}\right)^{-\mathfrak{b}}\right)\\ &\le \frac{1}{(t+T_0)^{\mathfrak{b}}}\frac{\mathfrak{b}}{t+T_0} = \frac{\mathfrak{b}}{(t+T_0)^{\mathfrak{b}+1}}.\end{aligned}$$

Here the inequality follows from the fact that $(1+x)^{-b}\ge 1-bx$, where $b\in(0,1)$ and $x>0$. Then,

$$\frac{\beta_t-\beta_{t+1}}{\beta_t^2} \le \frac{\mathfrak{b}}{(t+T_0)^{1-\mathfrak{b}}} \le \frac{1}{2},$$

where the final inequality follows from our assumption that $T_0\ge(2\mathfrak{b})^{1/(1-\mathfrak{b})}$. This implies that $\beta_t-\beta_{t+1}\le\frac{1}{2}\beta_t^2$ for all $t\ge 0$.

Now, suppose that $S_t\le 2\beta_t$ for some $t\ge 0$. Then,

$$S_{t+1} = (1-\beta_t)S_t+\beta_t^2 \le 2\beta_t(1-\beta_t)+\beta_t^2 = 2\beta_t-\beta_t^2 \le 2\beta_{t+1}.$$

Note that $S_0\le 2\beta_0$ and hence, by induction, $S_t\le 2\beta_t$ for all $t\ge 0$. This completes the proof for Lemma 4(a) with $\Gamma_3 = 2D_2$.

(b) Bound on $\sum_{i=0}^{t} \beta_i \mathbb{E}\left[\|\mathbf{v}(\mathbf{z}_i)\|^2\right]$. Using $(1/\lambda)$-Lipschitz property of the map $\mathbf{v}(\cdot)$ (7), we get

$$\begin{aligned}
\|\mathbf{v}(\mathbf{z}_t)\| &\leq \|\mathbf{v}(\mathbf{x}_t)\| + \|\mathbf{v}(\mathbf{z}_t) - \mathbf{v}(\mathbf{x}_t)\| \\
&\leq \|\mathbf{v}(\mathbf{x}_t)\| + \frac{1}{\lambda}\|\mathbf{z}_t - \mathbf{x}_t\| \\
&= \|\mathbf{v}(\mathbf{x}_t)\| + \frac{1}{\lambda}\|\mathbf{U}_t\| \\
\implies \|\mathbf{v}(\mathbf{z}_t)\|^2 &\leq 2\|\mathbf{v}(\mathbf{x}_t)\|^2 + \frac{2}{\lambda^2}\|\mathbf{U}_t\|^2.
\end{aligned}$$

Then using Lemma 4(a),

$$\begin{aligned}
\sum_{i=0}^{t} \beta_i \mathbb{E}\left[\|\mathbf{v}(\mathbf{z}_i)\|^2\right] &\leq 2\sum_{i=0}^{t} \beta_i \mathbb{E}\left[\|\mathbf{v}(\mathbf{x}_i)\|^2\right] + \frac{2}{\lambda^2}\sum_{i=0}^{t} \beta_i \mathbb{E}\left[\|\mathbf{U}_i\|^2\right] \\
&\leq 2\Gamma_2 + \frac{2\Gamma_3}{\lambda^2}\sum_{i=0}^{t} \beta_i^2 \\
&\leq 2\Gamma_2 + \frac{2\Gamma_3}{\lambda^2} D_1.
\end{aligned}$$

This completes the proof for part (b) with $\Gamma_4 = 2\Gamma_2 + 2\Gamma_3 D_1/\lambda^2$.

(c) Bound on $\mathbb{E}[\|\mathbf{v}(\mathbf{z}_i)\|^2]$ in terms of $\mathbb{E}[\|\mathbf{v}(\mathbf{z}_t)\|^2]$.

From the recursions of $\mathbf{x}_t$ and $\mathbf{U}_t$,

$$\begin{aligned}
\mathbf{z}_{t+1} &= \mathbf{x}_{t+1} - \mathbf{U}_{t+1} \\
&= \mathbf{x}_t + \beta_t(\mathbf{v}(\mathbf{x}_t) + \mathbf{M}_{t+1}) - ((1-\beta_t)\mathbf{U}_t + \beta_t \mathbf{M}_{t+1}) \\
&= \mathbf{z}_t + \beta_t(\mathbf{v}(\mathbf{x}_t) + \mathbf{U}_t) \\
&= \mathbf{z}_t + \beta_t(\mathbf{v}(\mathbf{z}_t) + \mathbf{e}_t),
\end{aligned}$$

where $\mathbf{e}_t \coloneqq \mathbf{v}(\mathbf{x}_t) - \mathbf{v}(\mathbf{z}_t) + \mathbf{U}_t$. Using (7), we have $\|\mathbf{e}_t\| \leq \|\mathbf{v}(\mathbf{x}_t) - \mathbf{v}(\mathbf{z}_t)\| + \|\mathbf{U}_t\| \leq (1 + 1/\lambda)\|\mathbf{U}_t\|$. This gives us

$$\implies \mathbb{E}\left[\|\mathbf{e}_t\|^2\right] \leq \left(1 + \frac{1}{\lambda}\right)^2 \mathbb{E}\left[\|\mathbf{U}_t\|^2\right] \leq \left(1 + \frac{1}{\lambda}\right)^2 \Gamma_3 \beta_t. \tag{8}$$

Next, note that

$$\begin{aligned}
\mathbf{z}_{t+1} - \mathbf{z}_t &= \beta_t(\mathbf{v}(\mathbf{z}_t) + \mathbf{e}_t) \\
\implies \mathbf{v}(\mathbf{z}_t) &= \frac{\mathbf{z}_{t+1} - \mathbf{z}_t - \beta_t \mathbf{e}_t}{\beta_t}.
\end{aligned}$$

Therefore,

$$\begin{aligned}
\|\mathbf{v}(\mathbf{z}_{t+1})\|^2 &= \|\mathbf{v}(\mathbf{z}_t)\|^2 + \|\mathbf{v}(\mathbf{z}_{t+1}) - \mathbf{v}(\mathbf{z}_t)\|^2 + 2\langle \mathbf{v}(\mathbf{z}_{t+1}) - \mathbf{v}(\mathbf{z}_t), \mathbf{v}(\mathbf{z}_t)\rangle \\
&= \|\mathbf{v}(\mathbf{z}_t)\|^2 + \|\mathbf{v}(\mathbf{z}_{t+1}) - \mathbf{v}(\mathbf{z}_t)\|^2 + \frac{2}{\beta_t}\langle \mathbf{v}(\mathbf{z}_{t+1}) - \mathbf{v}(\mathbf{z}_t), \mathbf{z}_{t+1} - \mathbf{z}_t - \beta_t \mathbf{e}_t\rangle \\
&\leq \|\mathbf{v}(\mathbf{z}_t)\|^2 + \|\mathbf{v}(\mathbf{z}_{t+1}) - \mathbf{v}(\mathbf{z}_t)\|^2 - \frac{2\lambda}{\beta_t}\|\mathbf{v}(\mathbf{z}_{t+1}) - \mathbf{v}(\mathbf{z}_t)\|^2 - 2\langle \mathbf{v}(\mathbf{z}_{t+1}) - \mathbf{v}(\mathbf{z}_t), \mathbf{e}_t\rangle \\
&= \|\mathbf{v}(\mathbf{z}_t)\|^2 - \left(\frac{2\lambda}{\beta_t} - 1\right)\|\mathbf{v}(\mathbf{z}_{t+1}) - \mathbf{v}(\mathbf{z}_t)\|^2 - 2\langle \mathbf{v}(\mathbf{z}_{t+1}) - \mathbf{v}(\mathbf{z}_t), \mathbf{e}_t\rangle.
\end{aligned} \tag{9}$$

Here the inequality follows from the fact that map $-\mathbf{v}(\cdot)$ is λ-co-coercive.

For any $c > 0$, we have,

$$0 \leq \left\|\sqrt{c}u + \frac{1}{\sqrt{c}}w\right\|^2 = c\|u\|^2 + 2\langle u, w\rangle + \frac{1}{c}\|w\|^2.$$

Hence,

$$-c\|u\|^2 - 2\langle u, w\rangle \leq \frac{1}{c}\|w\|^2$$

Taking $c = \frac{2\lambda}{\beta_t} - 1$, $u = \mathbf{v}(\mathbf{z}_{t+1}) - \mathbf{v}(\mathbf{z}_t)$, and $w = \mathbf{e}_t$,

$$-\left(\frac{2\lambda}{\beta_t} - 1\right)\|\mathbf{v}(\mathbf{z}_{t+1}) - \mathbf{v}(\mathbf{z}_t)\|^2 - 2\langle \mathbf{v}(\mathbf{z}_{t+1}) - \mathbf{v}(\mathbf{z}_t), \mathbf{e}_t\rangle \leq \frac{1}{\left(\frac{2\lambda}{\beta_t} - 1\right)}\|\mathbf{e}_t\|^2 \leq \frac{\beta_t}{\lambda}\|\mathbf{e}_t\|^2.$$

Here the final inequality follows from the assumption that $\beta_t \leq \lambda$ for all $t \geq 0$. Combining this with (9), we get

$$\|\mathbf{v}(\mathbf{z}_{t+1})\|^2 \leq \|\mathbf{v}(\mathbf{z}_t)\|^2 + \frac{\beta_t}{\lambda}\|\mathbf{e}_t\|^2.$$

This implies that

$$\mathbb{E}\left[\|\mathbf{v}(\mathbf{z}_{t+1})\|^2\right] \leq \mathbb{E}\left[\|\mathbf{v}(\mathbf{z}_t)\|^2\right] + \Gamma_5\beta_t^2,$$

where $\Gamma_5 = (1+1/\lambda)^2\Gamma_3/\lambda$. This follows from the bound on $\mathbb{E}[\|\mathbf{e}_t\|^2]$ in (8). Here, the second inequality uses $\beta_t \leq \lambda$, and the last inequality uses (8).

Iterating from $j = i$ to $t-1$ gives

$$\mathbb{E}\left[\|\mathbf{v}(\mathbf{z}_i)\|^2\right] \geq \mathbb{E}\left[\|\mathbf{v}(\mathbf{z}_t)\|^2\right] - \Gamma_5\sum_{j=i}^{t-1}\beta_j^2.$$

This completes the proof for Lemma 4. □

*E. **Proof for Lemma 5***

Proof. For each $i \leq t-1$, Lemma 4(c) gives

$$\mathbb{E}\left[\|\mathbf{v}(\mathbf{z}_t)\|^2\right] \leq \mathbb{E}\left[\|\mathbf{v}(\mathbf{z}_i)\|^2\right] + \Gamma_5\sum_{j=i}^{t-1}\beta_j^2.$$

Multiplying by β_i and summing from $i = 0$ to $t-1$,

$$\mathbb{E}\left[\|\mathbf{v}(\mathbf{z}_t)\|^2\right]\sum_{i=0}^{t-1}\beta_i \leq \sum_{i=0}^{t-1}\beta_i\mathbb{E}\left[\|\mathbf{v}(\mathbf{z}_i)\|^2\right] + \Gamma_5\sum_{i=0}^{t-1}\beta_i\sum_{j=i}^{t-1}\beta_j^2.$$

Adding $\beta_t\mathbb{E}\left[\|\mathbf{v}(\mathbf{z}_t)\|^2\right]$ to both sides and using Lemma 4(b),

$$\begin{aligned}
&\mathbb{E}\left[\|\mathbf{v}(\mathbf{z}_t)\|^2\right]\sum_{i=0}^{t}\beta_i \leq \Gamma_4 + \Gamma_5\sum_{i=0}^{t-1}\beta_i\sum_{j=i}^{t-1}\beta_j^2 \\
&\implies \mathbb{E}\left[\|\mathbf{v}(\mathbf{z}_t)\|^2\right] \leq \frac{\Gamma_4}{\sum_{i=0}^{t}\beta_i} + \Gamma_5\frac{\sum_{i=0}^{t-1}\beta_i\sum_{j=i}^{t-1}\beta_j^2}{\sum_{i=0}^{t}\beta_i}.
\end{aligned}$$

To bound the first term, we use the same simplification as the proof for Theorem 2 to get

$$\frac{\Gamma_4}{\sum_{i=0}^{t}\beta_i} \leq \frac{\Gamma_4 T_0^{\mathfrak{b}}}{(t+1)^{1-\mathfrak{b}}}.$$

Now, to bound the second term, note that

$$\sum_{i=0}^{j}\beta_i \leq \sum_{i=0}^{j}(i+1)^{-\mathfrak{b}} \leq 1 + \int_1^{j+1} x^{-\mathfrak{b}}dx \leq \frac{(j+1)^{1-\mathfrak{b}}}{1-\mathfrak{b}}.$$

Now, by swapping the order of summation and using the above bound, we get

$$\sum_{i=0}^{t-1}\beta_i\sum_{j=i}^{t-1}\beta_j^2 = \sum_{j=0}^{t-1}\left(\sum_{i=0}^{j}\beta_i\right)\beta_j^2 \leq \frac{1}{1-\mathfrak{b}}\sum_{j=0}^{t-1}(j+1)^{1-3\mathfrak{b}}.$$

Using standard integral comparison, we bound the above sum as

$$\sum_{j=0}^{t-1}(j+1)^{1-3\mathfrak{b}} \leq \begin{cases} \dfrac{(t+1)^{2-3\mathfrak{b}}}{2-3\mathfrak{b}}, & \mathfrak{b} \in (\frac{1}{2}, \frac{2}{3}), \\ 2\log(t+1), & \mathfrak{b} = \frac{2}{3}, \\ \displaystyle\sum_{j=0}^{\infty}(j+1)^{1-3\mathfrak{b}}, & \mathfrak{b} \in (\frac{2}{3}, 1). \end{cases}$$

Substituting this and using the lower bound on $\sum_{i=0}^{t} \beta_i$, we get

$$\Gamma_5 \frac{\sum_{i=0}^{t-1} \beta_i \sum_{j=i}^{t-1} \beta_j^2}{\sum_{i=0}^{t} \beta_i} \leq \begin{cases} \dfrac{\Gamma_5 T_0^{\mathfrak{b}}}{(1-\mathfrak{b})(2-3\mathfrak{b})}(t+1)^{-(2\mathfrak{b}-1)}, & \mathfrak{b} \in (\frac{1}{2}, \frac{2}{3}), \\ 6\Gamma_5 T_0^{2/3} \dfrac{\log(t+1)}{(t+1)^{1/3}}, & \mathfrak{b} = \frac{2}{3}, \\ \dfrac{\Gamma_5 T_0^{\mathfrak{b}}}{1-\mathfrak{b}} \left(\sum_{j=0}^{\infty} (j+1)^{1-3\mathfrak{b}}\right)(t+1)^{-(1-\mathfrak{b})}, & \mathfrak{b} \in (\frac{2}{3}, 1). \end{cases}$$

Using the upper bound on the first term and the above upper bound on the second term, we obtain

$$\mathbb{E}\left[\|\mathbf{v}(\mathbf{z}_t)\|^2\right] \leq \begin{cases} \Gamma_6 (t+1)^{-(2\mathfrak{b}-1)}, & \mathfrak{b} \in (\frac{1}{2}, \frac{2}{3}), \\ \Gamma_7 \dfrac{\log(t+1)}{(t+1)^{1/3}}, & \mathfrak{b} = \frac{2}{3}, \\ \Gamma_8 (t+1)^{-(1-\mathfrak{b})}, & \mathfrak{b} \in (\frac{2}{3}, 1). \end{cases}$$

Here

$$\Gamma_6 \coloneqq \Gamma_4 T_0^{\mathfrak{b}} + \frac{\Gamma_5 T_0^{\mathfrak{b}}}{(1-\mathfrak{b})(2-3\mathfrak{b})},$$
$$\Gamma_7 \coloneqq \frac{\Gamma_4 T_0^{2/3}}{\log 2} + 6\Gamma_5 T_0^{2/3},$$
$$\Gamma_8 \coloneqq \Gamma_4 T_0^{\mathfrak{b}} + \frac{\Gamma_5 T_0^{\mathfrak{b}}}{1-\mathfrak{b}} \sum_{j=0}^{\infty} (j+1)^{1-3\mathfrak{b}}.$$

This completes the proof for Lemma 5. □